Intestinal Obstruction: Colo-rectal Cancer

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Questions

- Does the patient have intestinal obstruction?
  - Ileus vs. mechanical obstruction
- What is the site and cause of obstruction?
  - Small bowel vs. large bowel
- Is the obstruction simple or strangulated?
- What investigations should be performed?
- Should the patient be treated conservatively and for how long?
- What are the indications for surgery?
- What are the surgical options?

Intestinal obstruction

- Mechanical obstruction implies a physical barrier to the aboral progress of intestinal contents

- Ileus implies failure of peristalsis to propel the intestinal contents with no mechanical barrier

Paralytic ileus:
causes

- Intra-peritoneal
  - Post-operative
  - Peritonitis or intra-abdominal abscess
  - Inflammatory/ infective conditions
  - Intestinal ischaemia

- Retroperitoneal
  - Retroperitoneal haematoma/infection
  - Aortic, spinal, urological operations
  - Pancreatitis

Paralytic ileus:
causes

Extra-abdominal conditions

- Metabolic abnormalities: electrolyte imbalance, sepsis, uraemia, hypothyroidism, lead poisoning, porphyria
- Medications: opiates, anticholinergics, antihistamines, catecholamines
- Spinal injury and operation

Paralytic ileus:
clinical features

- Abdominal distention
- Constipation
- Vomiting
- Abdominal pain: diffuse, constant and less severe
- Sluggish or absent bowel sounds

- Clinical features associated with the cause

Management

- Nil by mouth
- Intravenous fluid
- Nasogastric decompression
- Identify and treat the predisposing cause
- Monitor and assessment by daily abdominal X-rays and physical examination

Mechanical bowel obstruction

Pathophysiology

- Proximal bowel distended with gas and fluid
- Hypersecretion and loss of fluid to extracellular space and peritoneal cavity
- Bacterial overgrowth
- Compromise of blood supply, leading to necrosis and perforation of bowel (accelerated in close loop or stangulating obstruction)

Classification

- Partial vs. complete obstruction
- Chronic vs. acute
- Simple obstruction- obstruction of lumen, usually at one point only
- Strangulating obstruction- blood supply to bowel impaired
- Close loop obstruction-lumen occlusion in at least 2 points

Clinical features:
mechanical obstruction

- Colicky abdominal pain
- Abdominal distention
- Vomiting and nausea
- Constipation
- Increased bowel sounds

- Severity of each symptom depends on the level of obstruction

Past History

- Previous episodes of bowel obstruction
- Previous abdominal or pelvic operation
- History of cancer or abdominal/pelvic radiation
- History of abdominal inflammatory condition

Physical examination

- Assessment of vital signs and hydration status
- Abdominal examination
  - Surgical scars
  - External hernias
  - Abdominal mass
  - Peritoneal signs
  - Auscultation

Laboratory tests

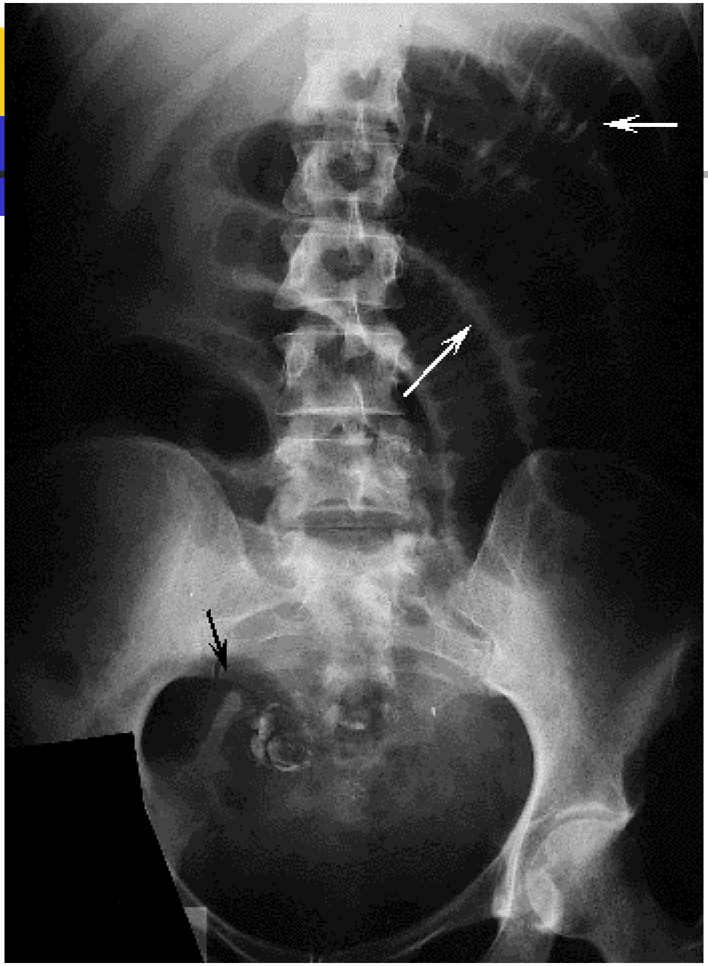
- Complete blood picture
- Serum electrolytes
- Arterial blood gas
- Renal function test
- Amylase

Imaging

- Chest X-rays
  - Exclude perforation with pneumoperitoneum

Imaging

- Abdominal X-rays (supine and erect films)
  - Are there dilated bowel loops?
  - Are air fluid levels present in erect film?
  - Any gas in the colon and the level of cut off?
  - Any evidence of strangulation: thumb printing, pneumatosis cystoides intestinalis, free peritoneal gas?
  - Any massive dilatation of colon?
  - Any air in the biliary tree?

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CT scan

- More sensitive than plain abdominal X-rays
- Level of obstruction (transition between dilated and collapsed loop)
- Lesions (tumor, foreign bodies)
- Viability of bowel (intravenous contrast)


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Contrast study

- Water soluble contrast
- Differentiates complete and partial obstruction
- ?therapeutic effect
- Barium study
  - Precipitates complete obstruction
  - Barium peritonitis

Causes of small bowel obstruction

Intraluminal

- Foreign bodies
- Gallstones
- Bezoars
- Worms

Intramural

- Tumor (primary or secondary)
- Strictures (Crohn's disease, radiation, anastomotic, drug induced)
- Intussusception

Causes of small bowel obstruction

- Extrinsic
  - Adhesions
  - Hernias
  - Volvulus
  - Intraperitoneal malignancy

Etiology of small bowel obstruction

Etiology	Number of Patients (%) (n = 552)	Number of Admissions (%) (n = 1,001)
Adhesions	410 (74)	675 (68)
Crohn's disease	39 (7)	114 (11)
Neoplasm	25 (5)	55 (5)
Hernia	12 (2)	28 (3)
Radiation injury	6 (1)	12 (1)
Combination of above*	20 (4)	56 (6)
Miscellaneous†	40 (7)	61 (7)

Miller G, et al, Am J Surg 2000

Management decision

Urgent surgery

vs.

conservative treatment

Indications for urgent surgery

- Incarcerated, strangulated hernia
- Suspected or proven strangulation
- Peritonitis
- Pneumoperitoneum
- Pneumatosis cystoides intestinalis
- Close loop obstruction
- Volvulus with peritoneal signs

Features suggestive of strangulation

- Increased abdominal pain and tenderness
- Blood in the vomitus
- Fever and increased white cell count
- Imaging:
  - Thumbprinting
  - loss of mucosal pattern
  - gas within the bowel wall or within intrahepatic branches of the portal vein may be seen in strangulation.

Non operative treatment

- For partial obstruction
  - Adhesions
  - Crohn's disease
  - Radiation stricture
  - Disseminated malignant disease

Non-operative treatment

- Intravenous fluid and electrolytes
- Nasogastric decompression
- Nutrition when prolonged fasting is anticipated
- Frequent monitor of vital signs, abdominal signs and X-rays

Resolution of obstruction

- Less abdominal distention
- Reduction of nasogastric output
- Passage of flatus and bowel movement
- Resolution in abdominal X-rays

- Unresolved obstruction ⟶ surgical treatment (duration of conservative treatment controversial, usually 48 hours)

Adhesive obstruction

- The most common cause of small bowel obstruction in western countries
- Adhesions:
  - Congenital
  - Post inflammation
  - Formed after abdominal surgery

Prevention of adhesions

- Gentle handling of bowel during surgery
- Removal of powder from gloves
- ?use saline lavage
- Sodium hyaluronate bioresorbable membrane (seprafilm)

Adhesive obstruction

- Clinical features of small bowel obstruction with previous abdominal surgery
- Success rate of non operative treatment: about 50%
- Indications for surgery
  - Non responsive to conservative treatment
  - Clinical features of strangulation

Adhesive obstruction

Controversies:

- Duration of conservative treatment
- Administration of water soluble contrast
  - Differentiates partial from complete obstruction
  - Therapeutic effect?
  - Reduced operating rate?
  - Shorten hospital stay


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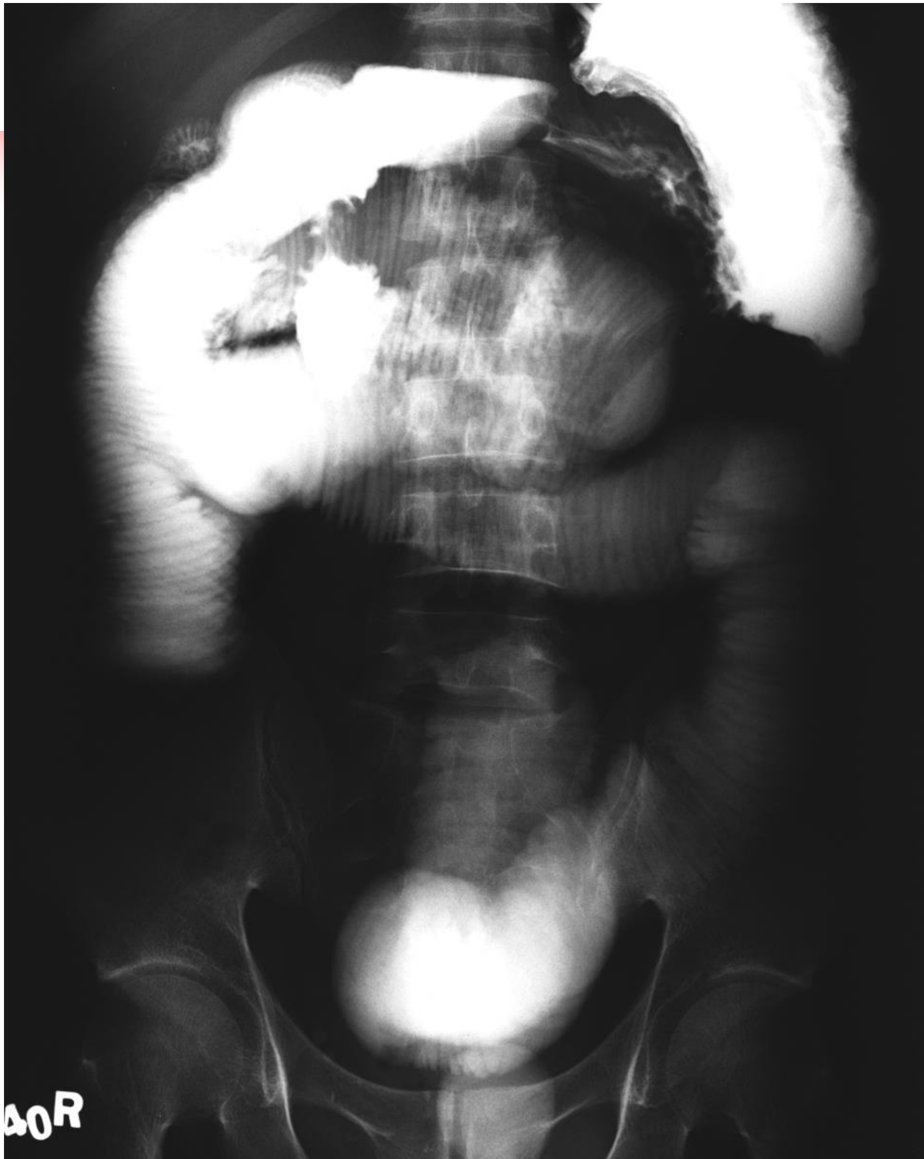
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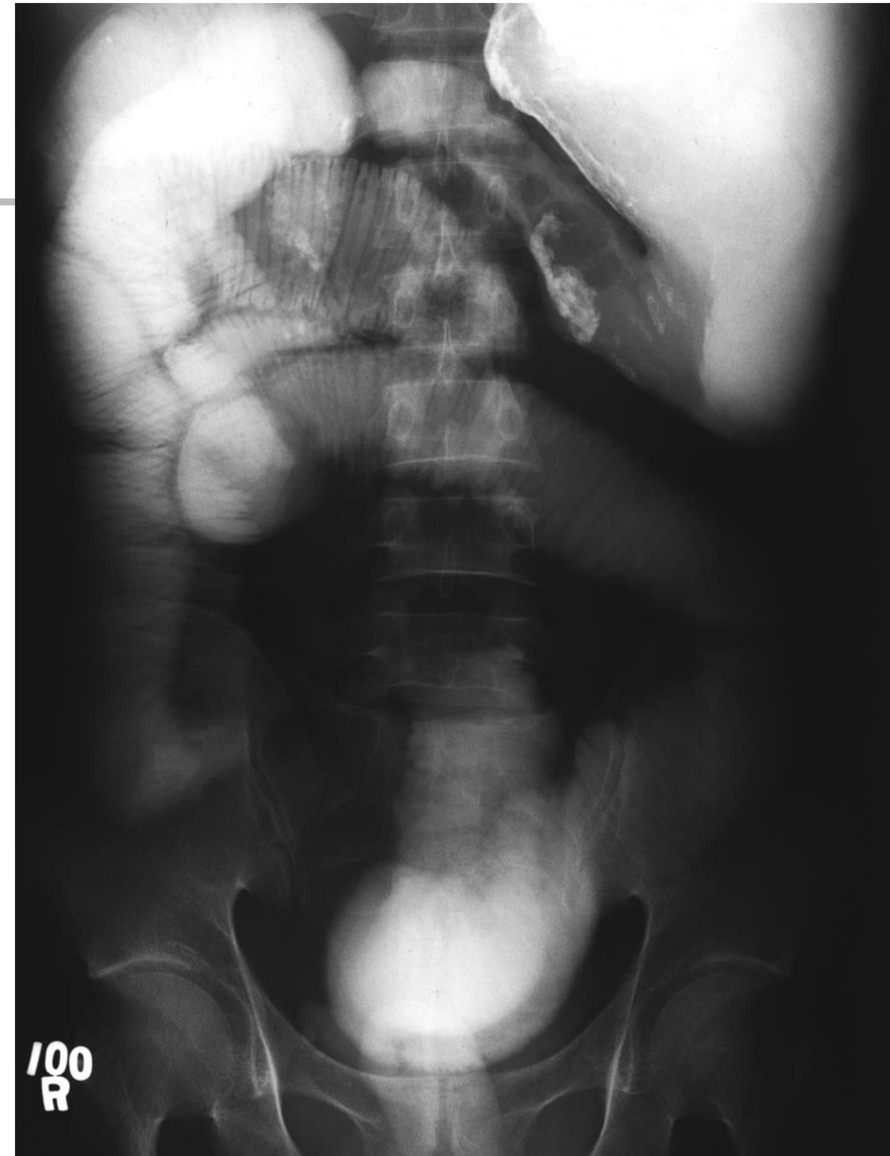
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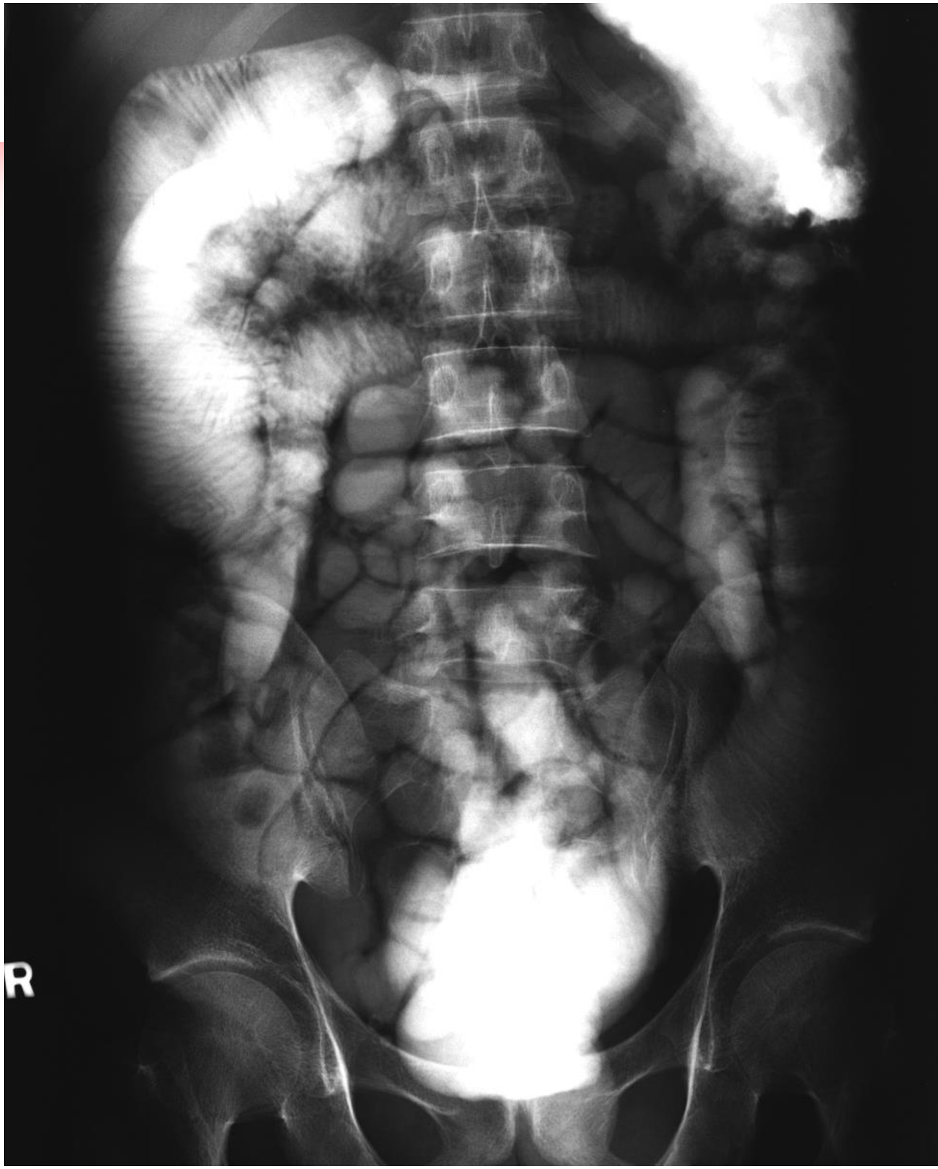
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Operative treatment: small bowel obstruction

- Enterolysis (lysis of adhesions and release of constricting bands)
- Repair of hernia
- Foreign bodies (Bezoars, gallstones)
  - Break down and push to colon
  - Enterotomy and removal
- Bowel resection
  - Strangulation with gangrenous bowel
  - Unhealthy bowel

Prognosis

- Mortality
  - Non strangulating obstruction: 2%
  - Strangulating obstruction: 10-30%

Colonic obstruction

- About 15% of intestinal obstruction
- Usually at sigmoid colon
- A lesion at ileocaecal valve presents as small bowel obstruction
- Competence of ileocaecal valve determines the clinical features of distal colon obstruction

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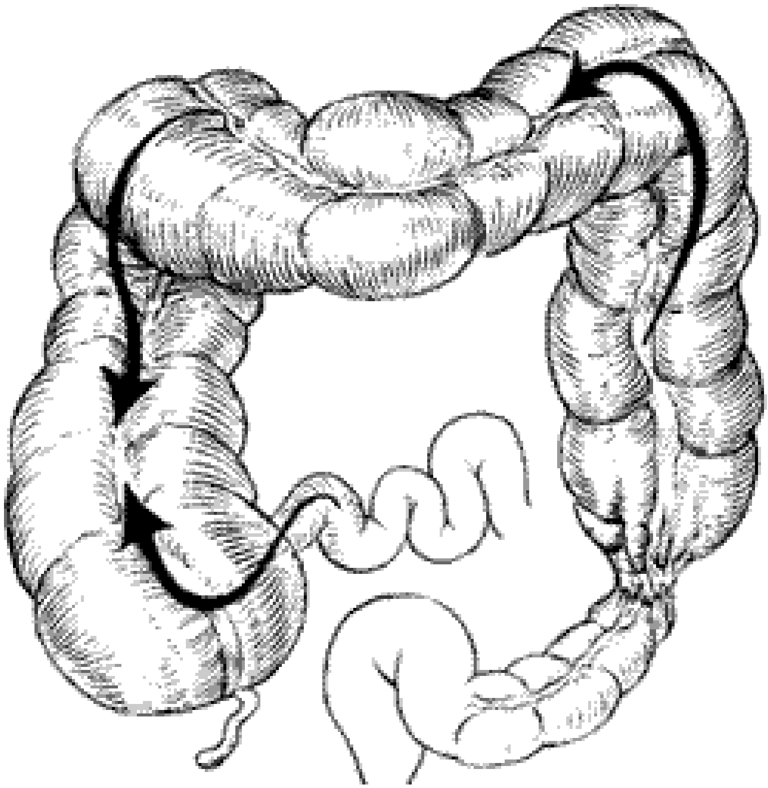

Competent ileocaecal valve

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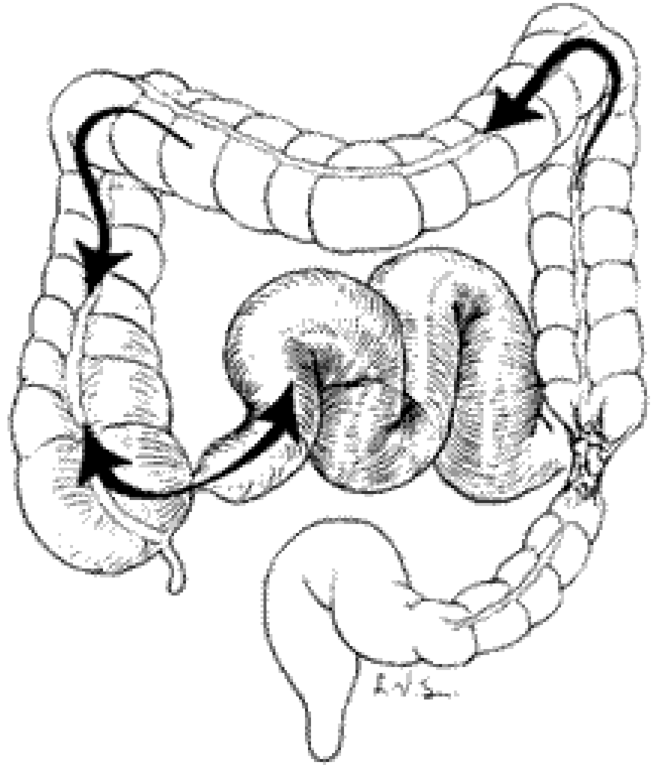


Incompetent ileocaecal valve

Causes of colonic obstruction

- Cancer
- Volvulus
- Diverticulitis
- Stricture: anastomotic, radiation, ischaemic, endometriotic
- Extrinsic compression: metastasis, pelvic/extraperitoneal tumor

Obstructing colorectal cancer

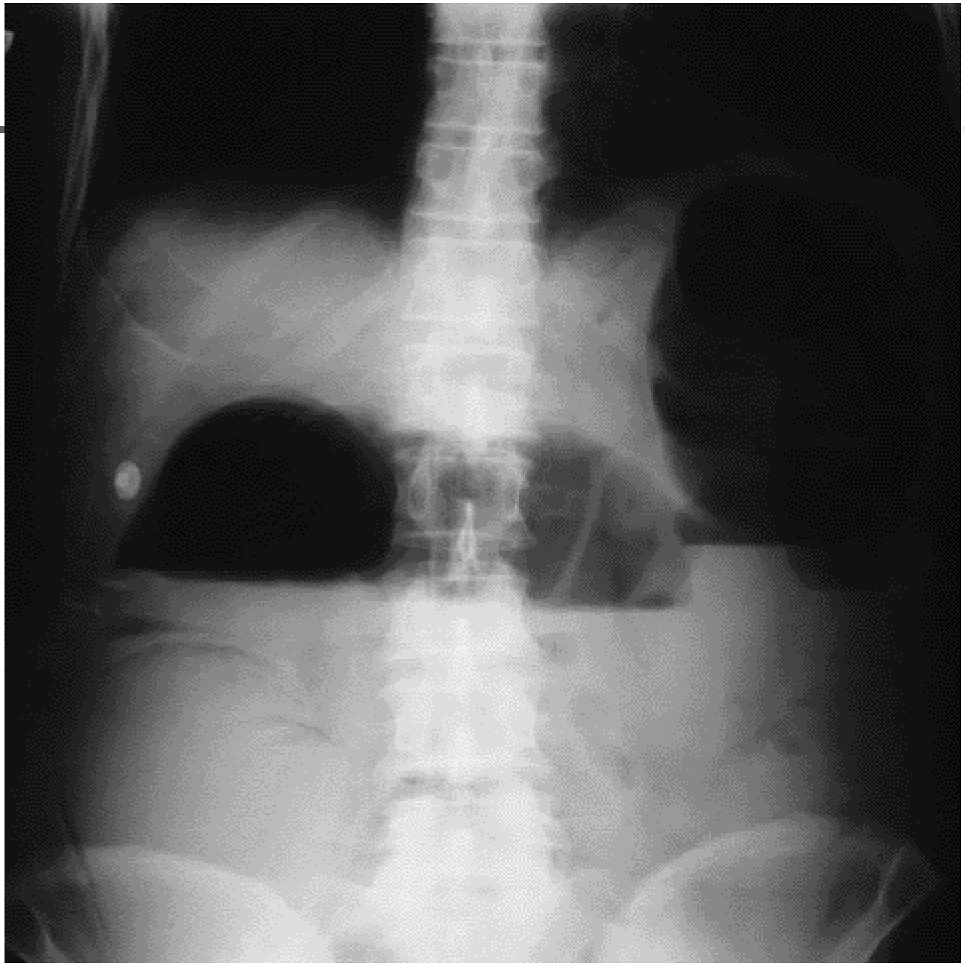
- 15-20% of patients with colorectal cancer present with intestinal obstruction
- Characteristics
  - More advanced cancer
  - Elderly patients with comorbidity
  - High operative mortality and morbidity
  - Worse prognosis

Management

- Resuscitation
- Diagnosis:
  - Clinical
  - Abdominal X-rays
  - CT scan
  - Sigmoidoscopy/colonoscopy
  - Contrast enema

Investigations

- Lower gastrointestinal endoscopy
  - Diagnostic
  - Therapeutic
    - Decompression in sigmoid volvulus and pseudo-obstruction
    - Stenting
  - Cautions: to avoid excessive insufflation of gas



CT scan

- Intravenous contrast, rectal contrast
- Site of obstruction (transition of dilated loop and collapsed loop)
- Mass lesion
- Perfusion of bowel wall
- Distant disease in case of malignancy

Operation

- Resection
  - Primary anastomosis
  - Without anastomosis
- Non resection
  - Proximal stoma
  - Bypass

Determinants of procedures

- Patient’s factors
  - general condition and nutritional status
  - haemodynamic status
  - ?sepsis
  - condition of the remaining bowel
- Tumor (lesion) factors
  - Site of lesion (right colon vs. left colon vs. rectum)
  - Invasion to adjacent structures
  - ?perforation/contamination of peritoneal cavity
- Surgeon’s factors
  - Experience in bowel resection and anastomosis in emergency

Operative treatment

Right-sided obstruction: caecum to splenic flexure

- Resection and anastomosis if the patient is stable (right or extended right colectomy)
- Resection without anastomosis if the patient or the bowel condition is not favourable
- Non resection

  Stoma or bypass (advanced tumor)

Left sided obstruction

- Factors to consider
  - Competence of ileocaecal valve (close loop obstruction $\longrightarrow$ perforation)
  - Heavy bacterial and faecal load in proximal colon
  - Edematous unhealthy proximal colon
  - Poor general condition of patient:
    - Malignancy and malnutrition
    - Dehydration

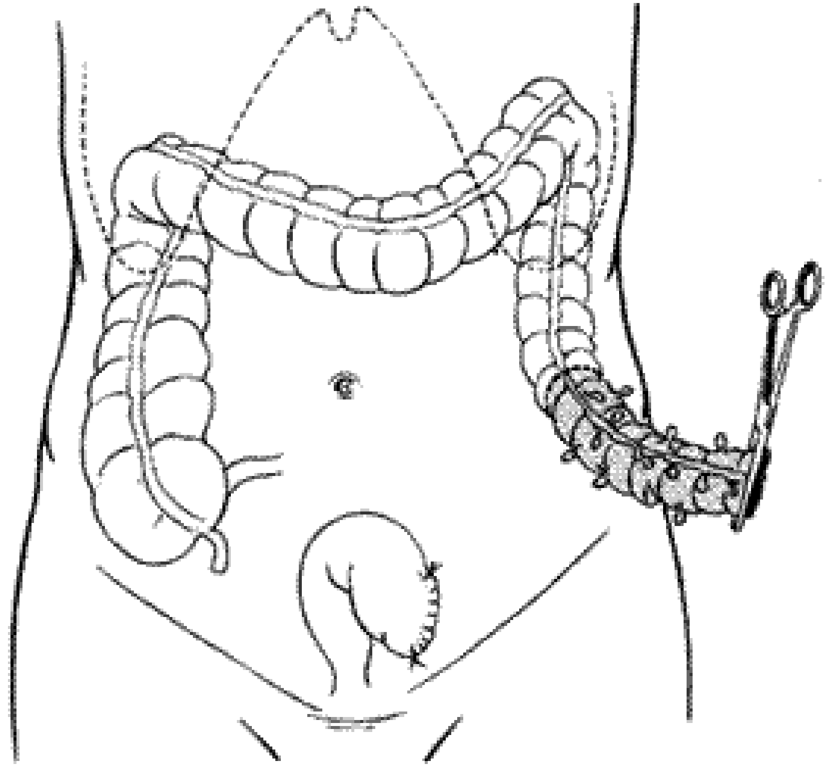
  - Primary anastomosis is risky

Left sided colonic obstruction

- 3-stage operation (not commonly performed today)
  - Transverse colostomy
  - Resection and anastomosis
  - Closure of colostomy

Left sided colonic obstruction

- Hartmann's operation
  - Resection without anastomosis



Left sided colonic obstruction

- Primary resection and anastomosis
  - Segmental resection with primary anastomosis (on table lavage)
  - Subtotal colectomy with anastomosis of ileum and distal colon/rectum

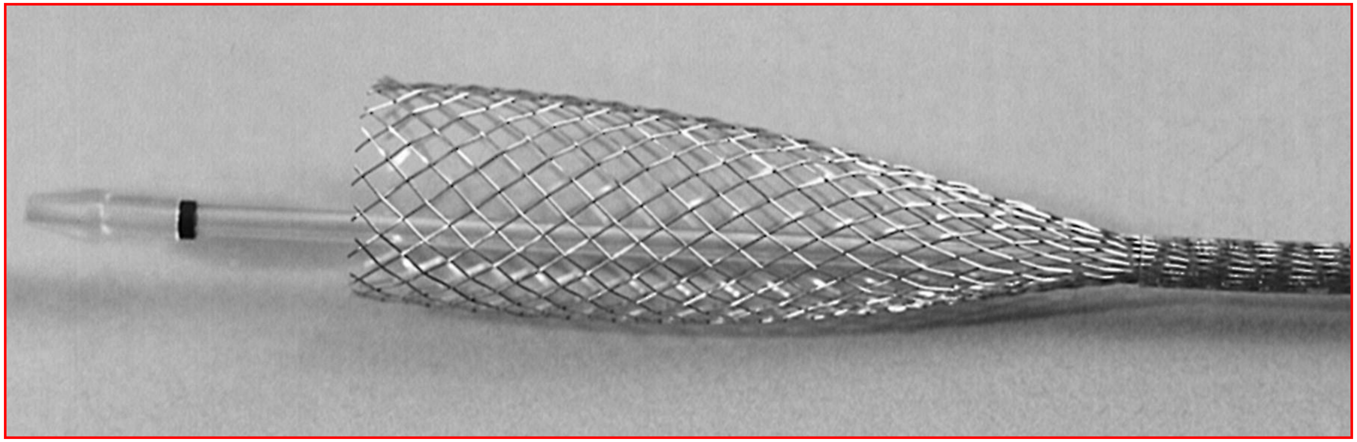
Prognosis of emergency surgery for colonic obstruction

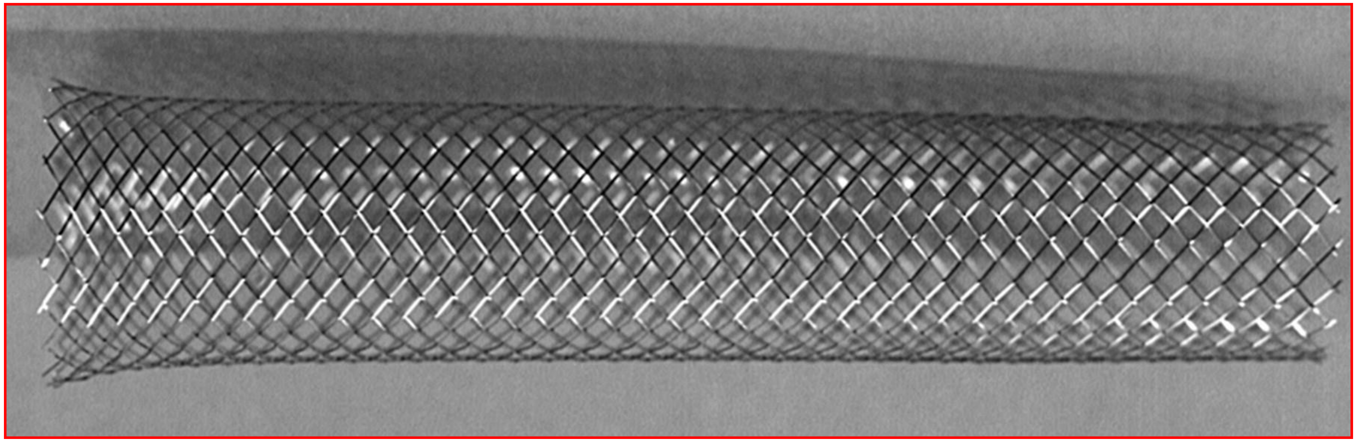
- Mortality: more than 10%
  - Comorbidity
  - Advanced malignancy

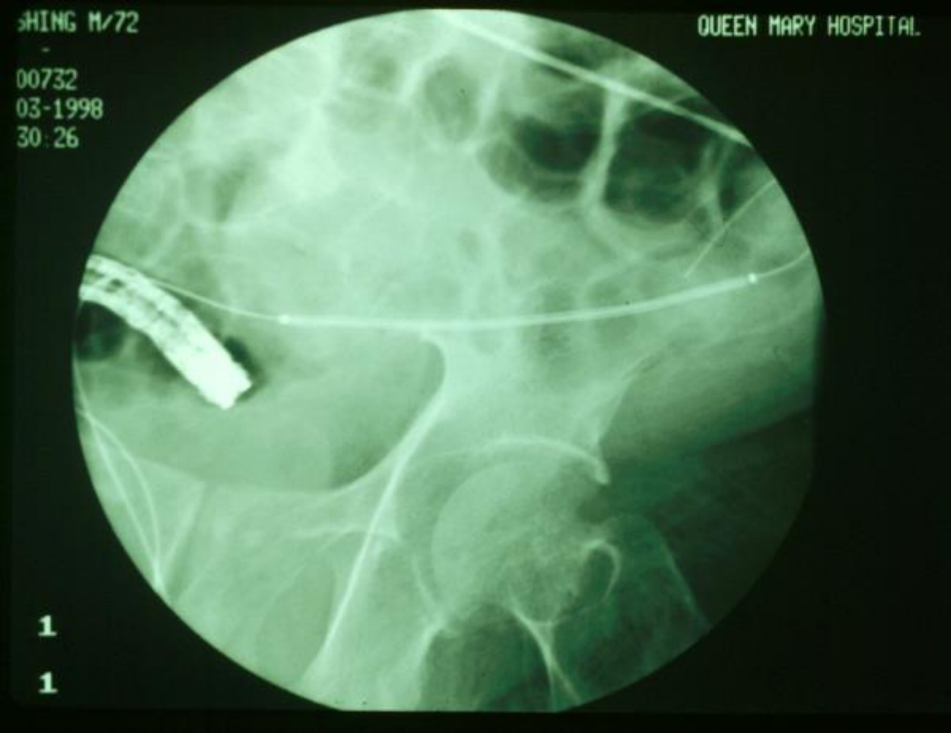
Non surgical treatment

- Insertion of metallic stent
  - Made of metal alloys
  - Self expanding mechanism
  - Insert and deploy under endoscopic and/or fluoroscopic guidance
  - For definitive palliation (unresectable, metastatic disease)
  - As a bridge to surgery

Expandable metallic stent



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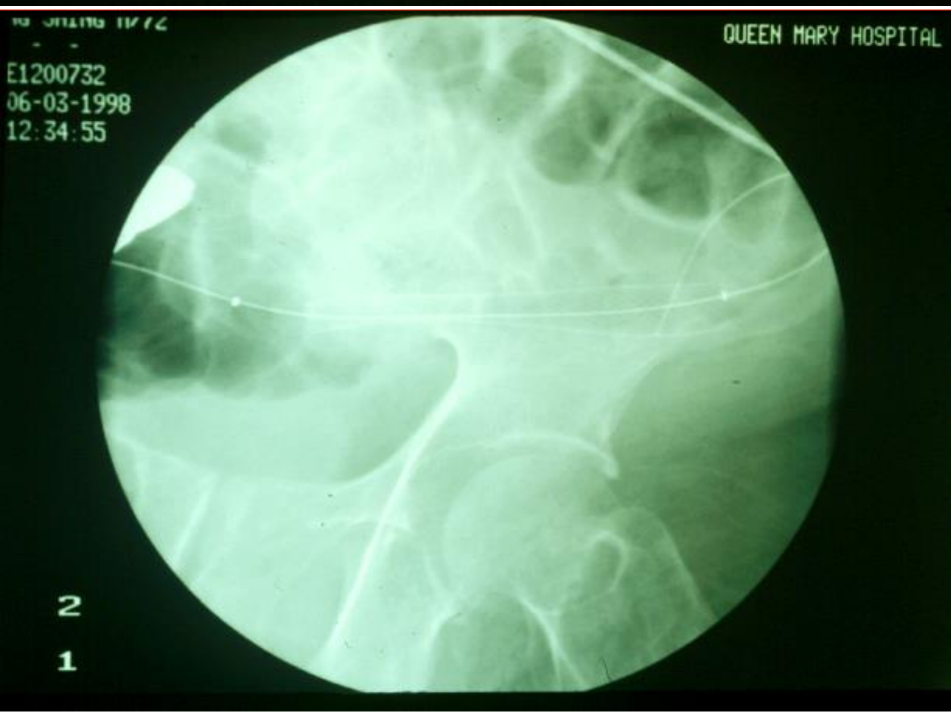
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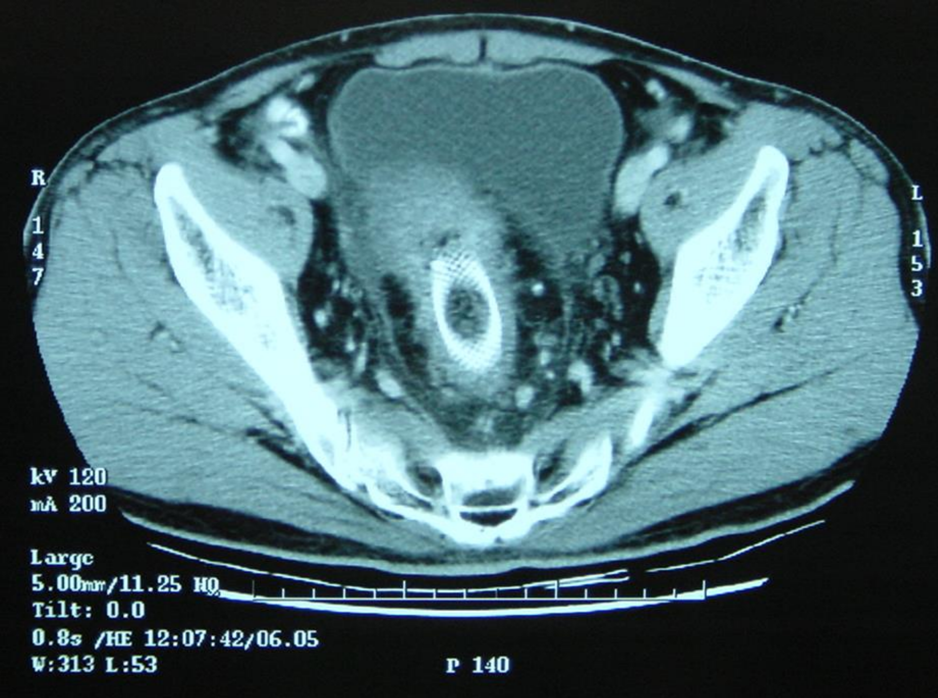
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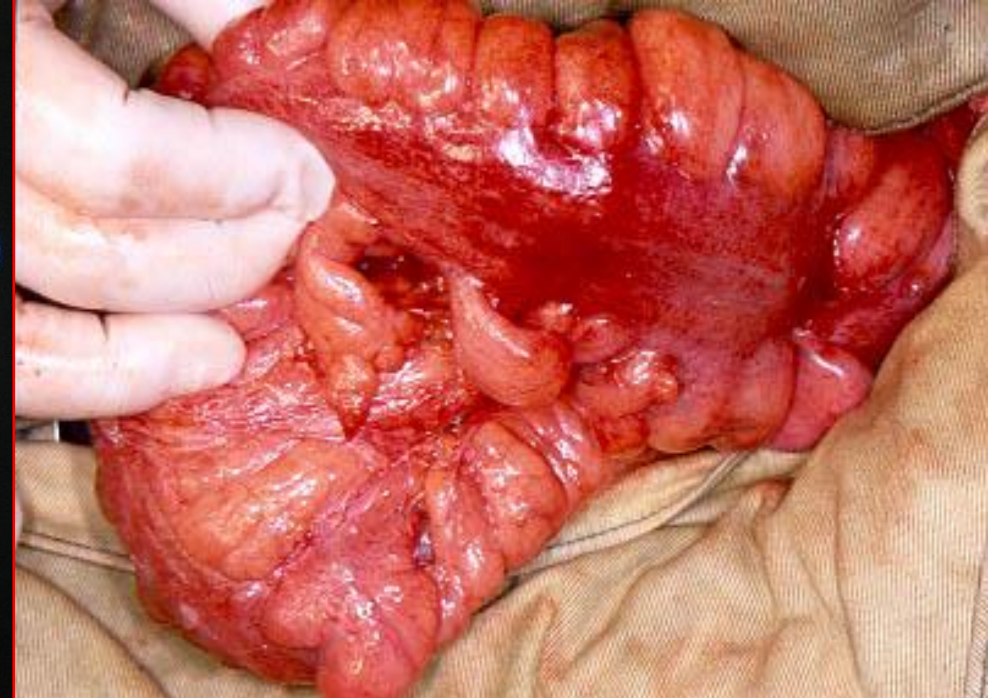


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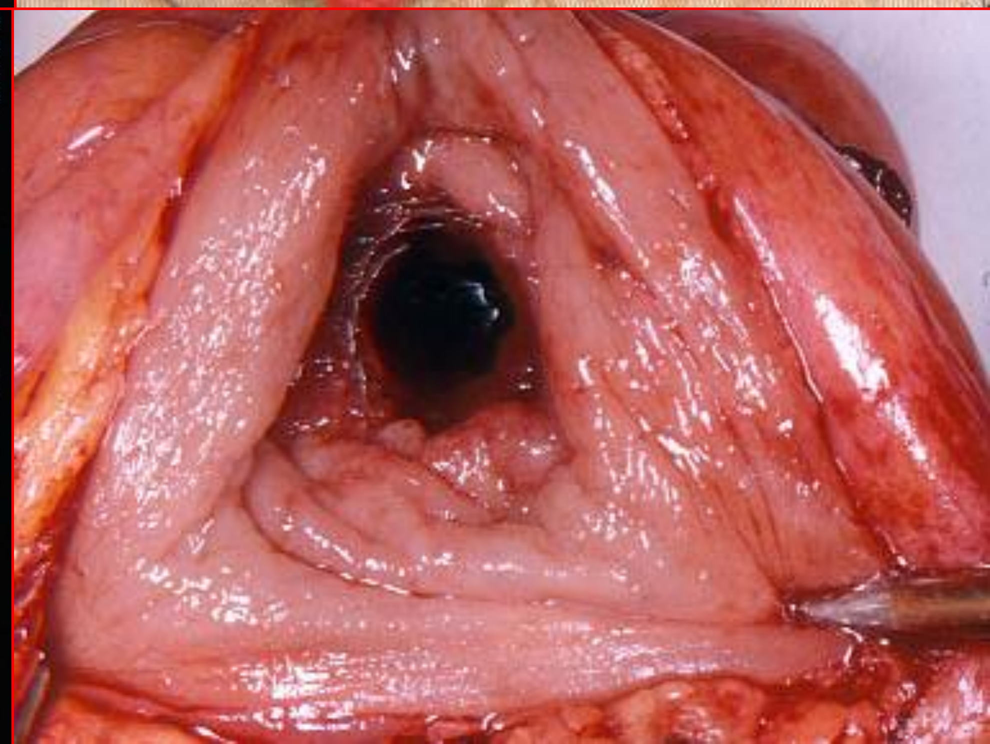


Stenting for colorectal malignancy

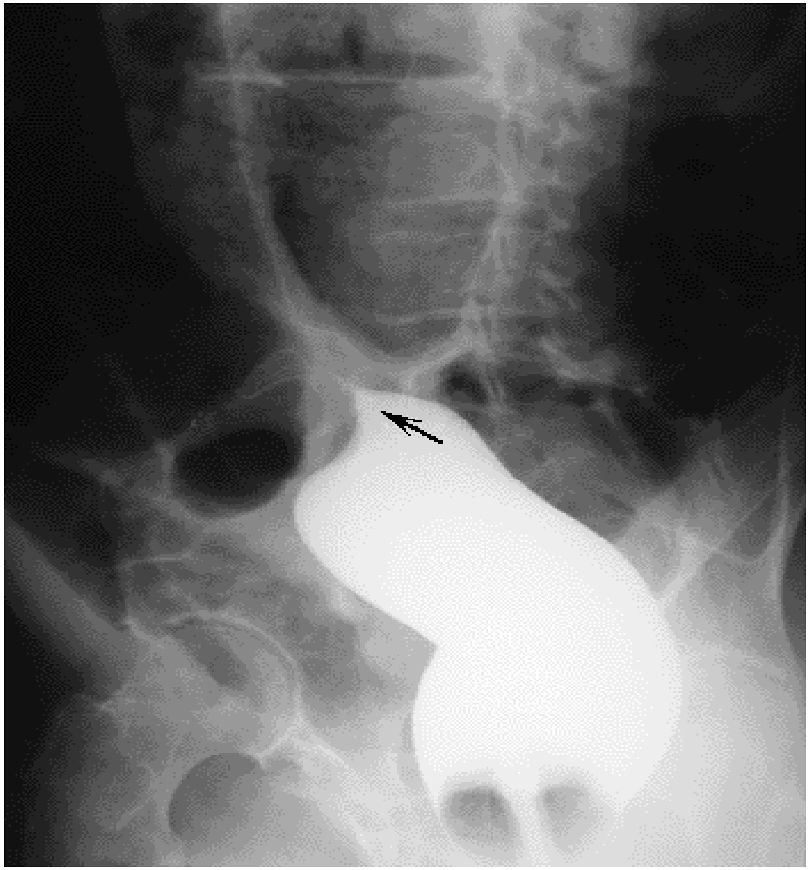
- Definitive palliation
  - Avoids surgery
  - Avoids stoma
- As a bridge to surgery
  - Avoids emergency surgery
  - Elective operation with bowel preparation
  - More time to stage the disease
  - Lower operative mortality and mortality
  - Reduces stoma rate

Volvulus of colon

- Rotation of the colon along the axis formed by its mesentery
- Colon obstruction with impairment of circulation
- Occurs commonly at sigmoid (65%) or caecum(30%)
- X-rays: dilated sigmoid (coffee bean)

Sigmoid volvulus

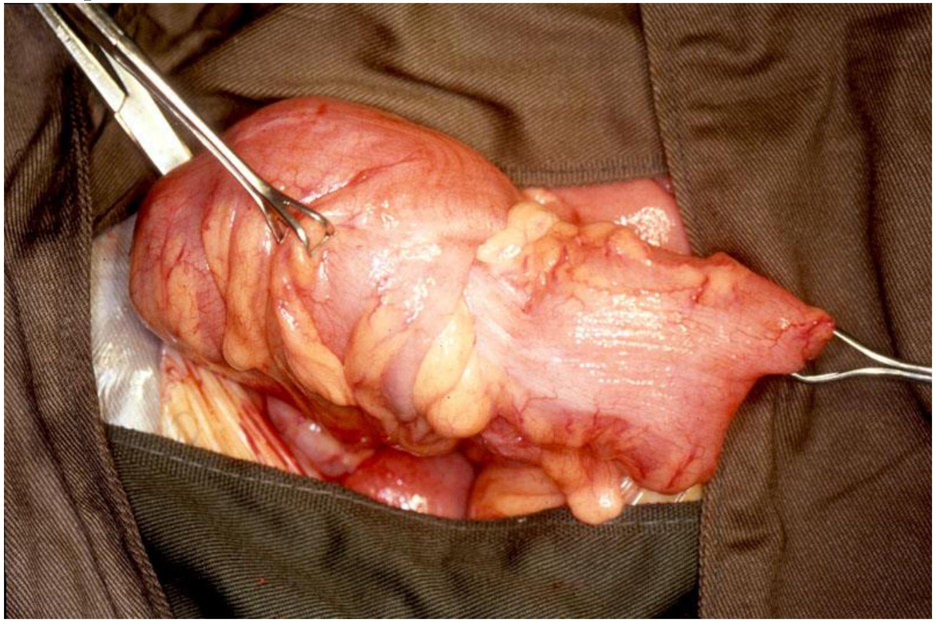
- Barium enema:
  - Bird's beak or ace of spade sign

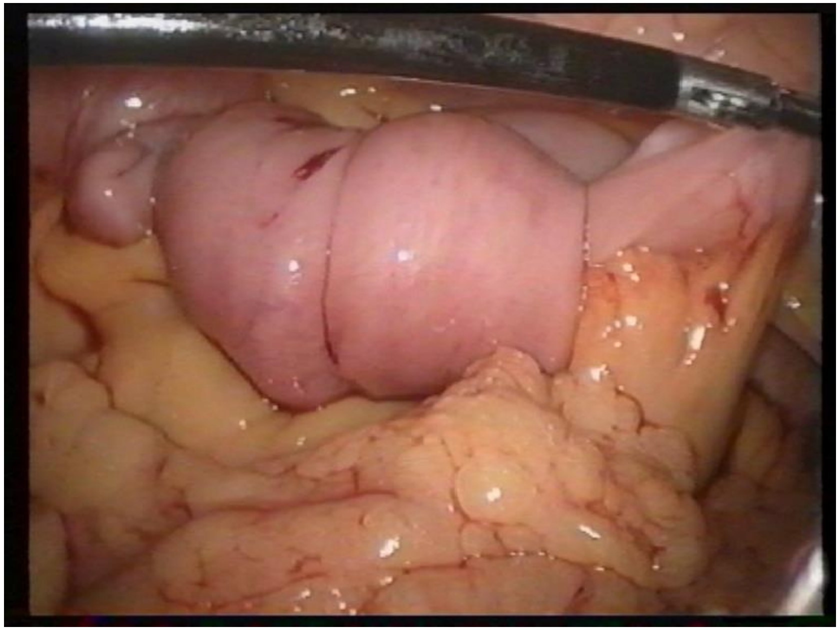


Volvulus

- Treatment
  - Sigmoidoscopic decompression (recurrence: 50%)
  - Surgery (perforation, strangulation or failed decompression)
    - Resection

Intussusception





Intussusception

- Intussusception in adults
  - A lesion is usually found as the leading point
  - Surgery is usually indicated

Pseudo-obstruction (Ogilvie's syndrome)

- Massive colon dilatation in the absence of mechanical obstruction
- Usually associated with bedridden patients with severe extracolonic diseases or trauma
- Distended abdomen without pain
- X-rays: severe gaseous distention of colon

Pseudo-obstruction

- Management
  - To exclude mechanical obstruction
  - Nasogastric tube feeding and enemas
  - Colonoscopic decompression
  - Rectal tube decompression
  - Neostigmine
  - Caecostomy

Summaries

- Intestinal obstruction is a common surgical emergency
- Management: initial resuscitation followed by determination of the site and cause of obstruction
- Decision on surgery and timing of surgery important
- High mortality if complications occur